

SHOU-TZU HAN

TRUSTWORTHY AI | LLM REASONING | AGENTIC RETRIEVAL

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Exploring postdoctoral research beginning in 2028.

ACADEMIC PROFILE

Computer Science Ph.D. researcher developing trustworthy and interpretable AI systems. Research spans mechanistic analysis of LLM reasoning, agentic retrieval for scientific discovery, and human-centered AI. Experience designing reproducible transformer experiments, quantitative evaluation frameworks, and research prototypes.

Research interests: Trustworthy AI; mechanistic interpretability; LLM robustness; agentic RAG; NLP; human-centered AI

EDUCATION & ACADEMIC APPOINTMENT

Wayne State University - Ph.D. in Computer Science 2026-Present

Graduate Research Assistant, Trustworthy AI Lab | Detroit, MI | Target completion: 2028

RESEARCH PREPRINTS

Han, S.-T., Rizk, R., & Santosh, K. C. (2026). Fragile Reasoning: A Mechanistic Analysis of LLM Sensitivity to Meaning-Preserving Perturbations. [arXiv:2604.01639](#).

Han, S.-T. (2026). Novelty-Aware Agentic Retrieval: Comparing Research Contributions Through Structured Multi-Step Reasoning. [arXiv:2606.22151](#).

Han, S.-T. (2025). Narrative-Centered Emotional Reflection: Scaffolding Autonomous Emotional Literacy with AI. [arXiv:2504.20342](#).

SELECTED RESEARCH CONTRIBUTIONS

LLM Robustness & Mechanistic Interpretability 2026

- Evaluated 3 open-weight LLMs on 677 paired GSM8K problems; developed logit-lens, activation-patching, component-ablation, and repair analyses that exposed 28.8%-45.1% answer-flip rates and achieved up to 12.2% recovery.

Novelty-Aware Agentic Retrieval 2026

- Built a 6-component agentic RAG system for comparative analysis of a 100-paper corpus; generated structured contribution and research-gap artifacts with main-query Precision@5 of 1.000 and nDCG@5 of 0.752.

Narrative-Centered Emotional AI 2025

- Developed an AI-guided reflection prototype combining emotion detection, layered prompts, and narrative generation; completed an early 12-participant pilot evaluation.

RESEARCH METHODS & LANGUAGES

Methods & tools: Python; Hugging Face Transformers; GPU/HPC experimentation; logit lens; activation patching; model ablation; retrieval evaluation

Languages: Mandarin Chinese (native); English